



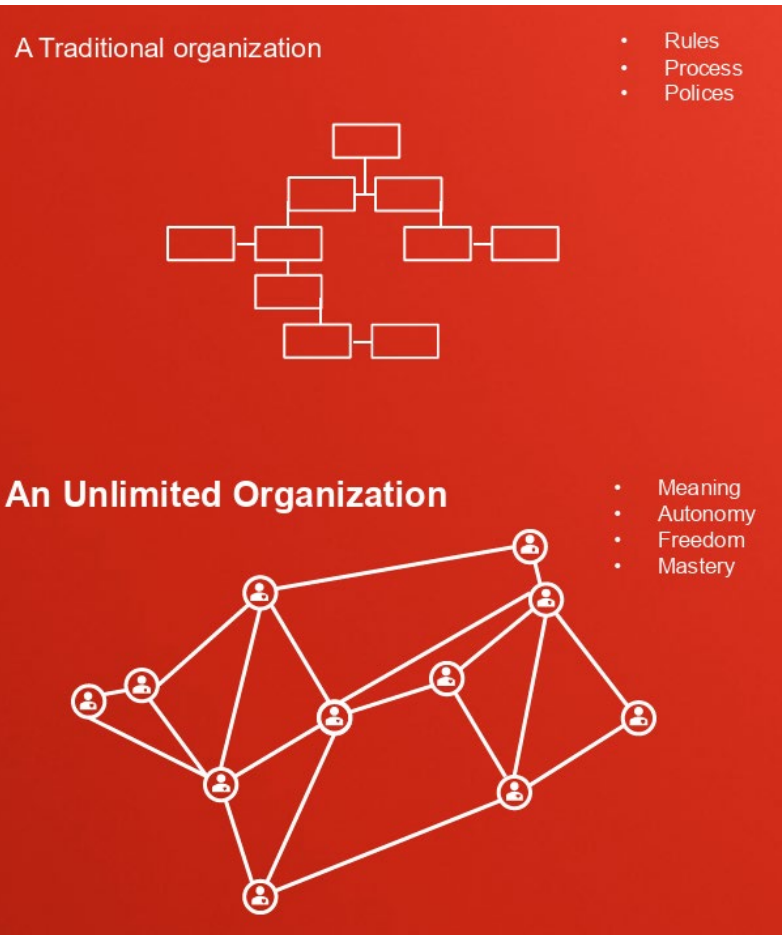
AGILE SOFTWARE ENGINEERING: IMPROVING AGILITY

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The Trackunit organization (p.18)



The Trackunit Product Development Life Cycle (p.29)

• 4 Core Phases

Customer-Problem Fit → Problem-Solution Fit → Product-Market Fit → End of Life.

Ensures customer needs are understood, validated, solved, and scaled sustainably

• 8 Stages

Define the “how” of delivery, guiding execution and tangible outcomes.

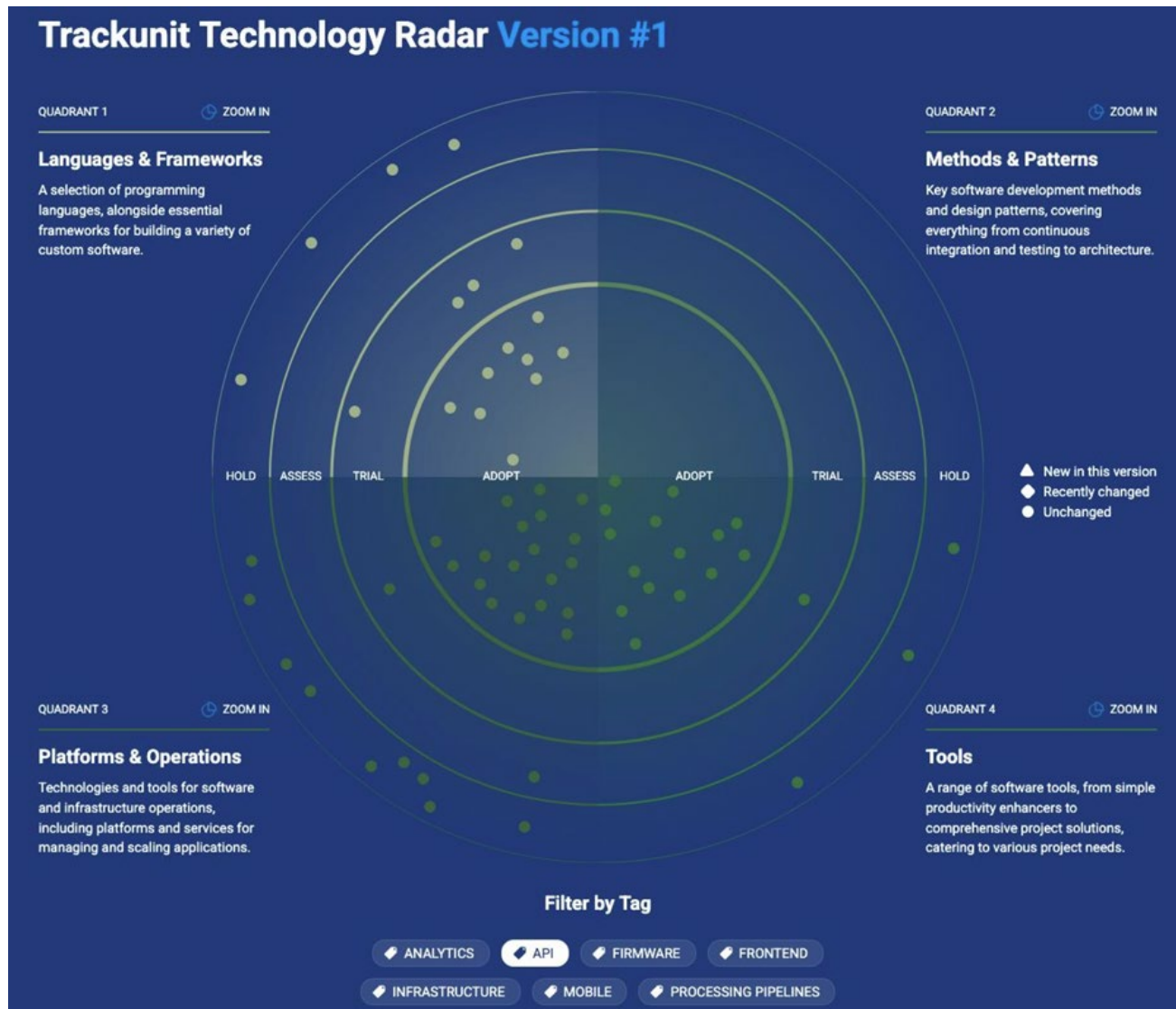
• Gates (Decision Checkpoints)

Formal review points between stages for go/no-go decisions.

Ensure accountability, alignment, and quality before progressing.

The Trackunit Software Development Life Cycle (p.31)







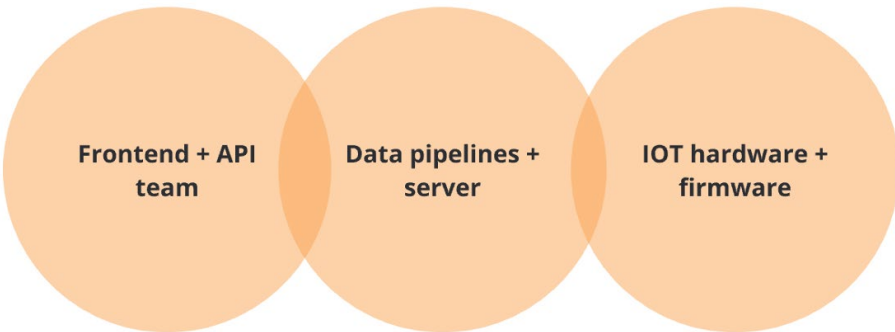
Golden paths

- Opinionated and supported path to building software
- Multiple golden paths exists
- Governed by communities and teams
- By choosing to follow the golden path, you will get things out of the box, such as backup, monitoring or other



2018

~ 25 people



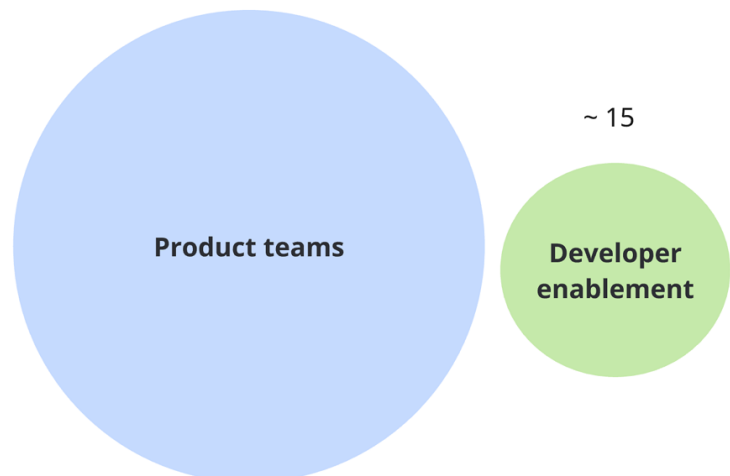
Frontend + API
team

Data pipelines +
server

IOT hardware +
firmware

Team constellation - 2025

~ 180



Product teams

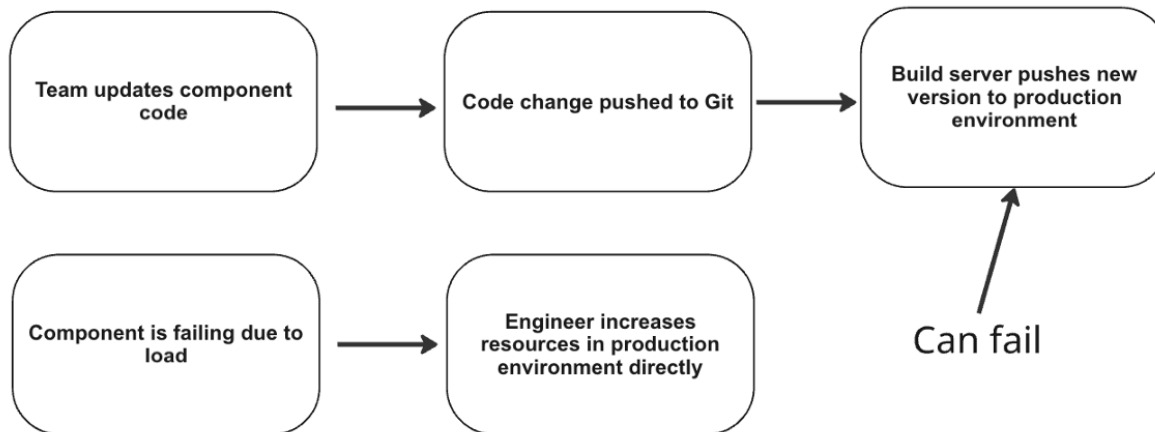
~ 15

Developer
enablement

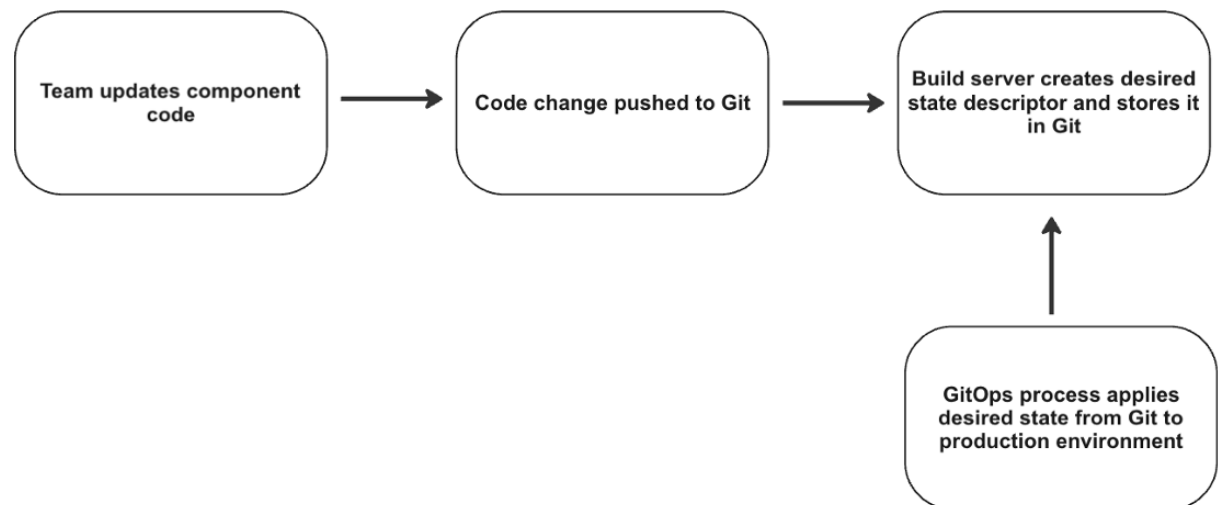




Before GitOps (p.66)



After GitOps (p.67)



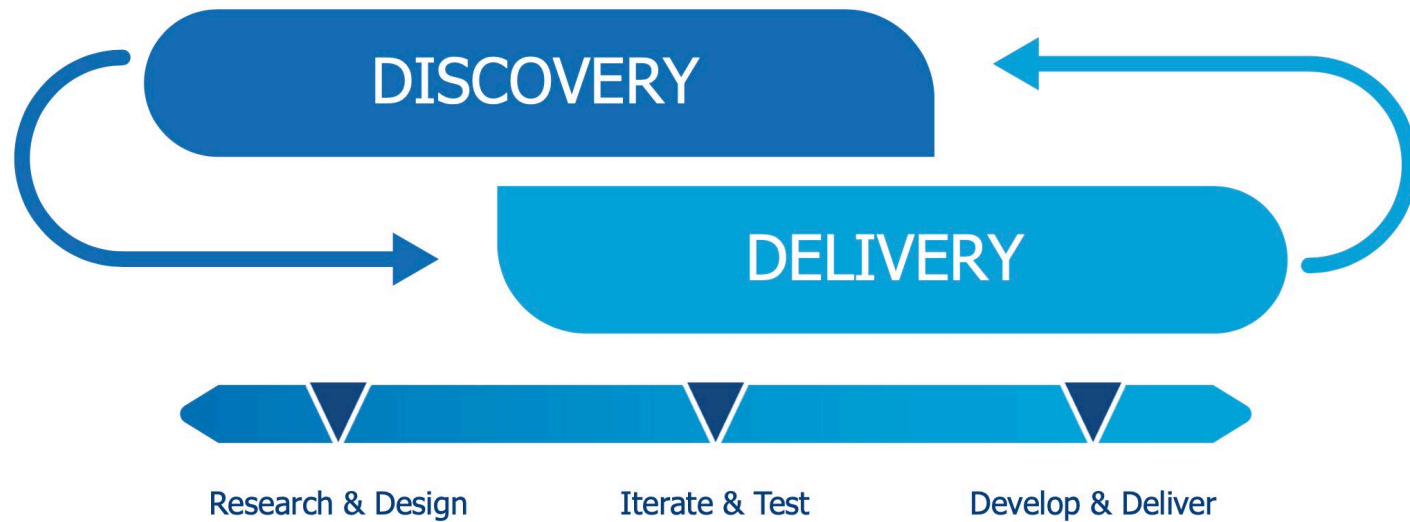
Lecture objectives

Knowledge about agile user-centered design and agility in an organizational context

Skills in defining and managing the agility of software processes

Competency to diagnose and improve agility in an organizational context

Dual Track Agile

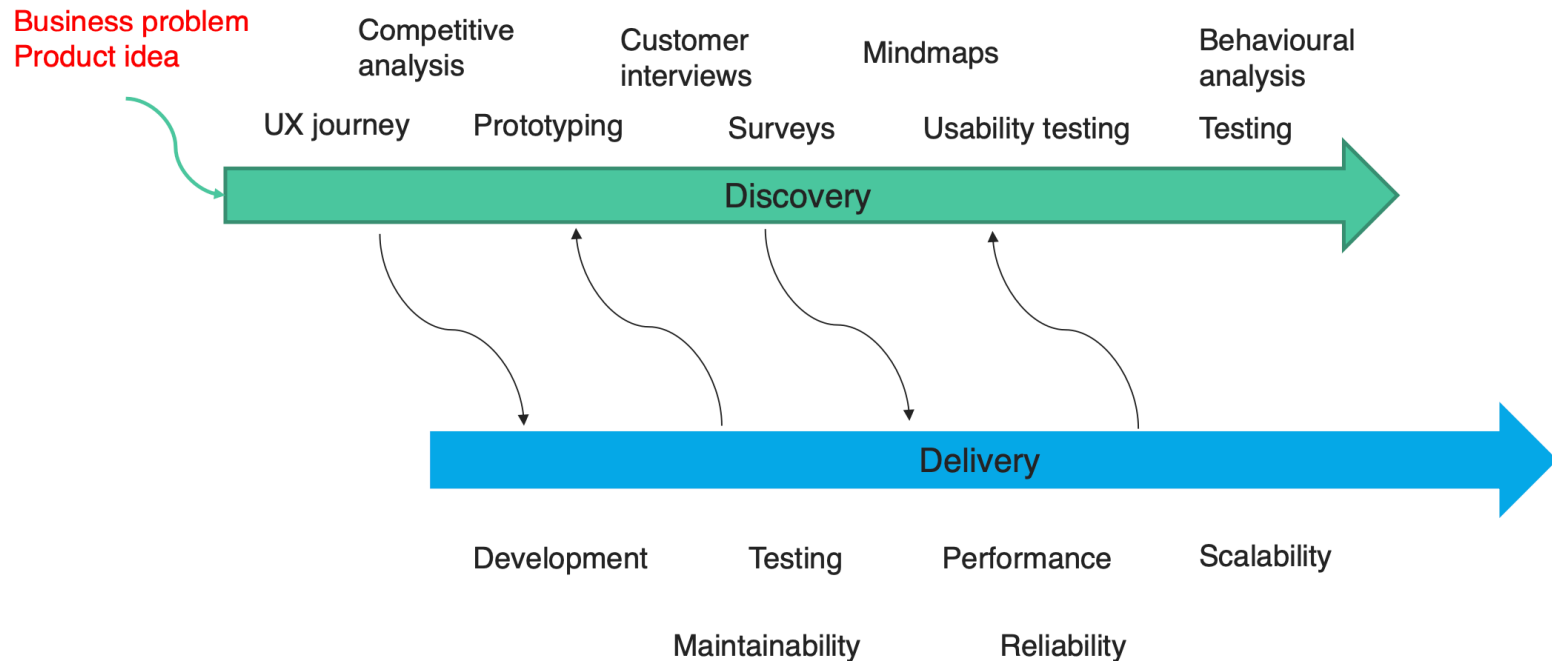


Dual Track Agile: Objectives

- ❖ Reduce the risk of complex and uncertain product ideas.
- ❖ A dedicated track to validate ideas/hypotheses, i.e., the discovery track.
- ❖ Validated ideas are converted to clear product backlog items.
- ❖ Teams can reduce the risk of developing the incorrect product or features that do not satisfy customer requirements or market demand by having a dedicated track for discovery and validation.
- ❖ The discovery track aids in the validation of ideas and the gathering of input early in the development process.
- ❖ It allows teams to make informed decisions and create valid, tested, and clear product backlog items for the delivery track.
- ❖ This lowers the risk of spending time and resources in developing a product that does not eventually satisfy the requirements of consumers or the market.



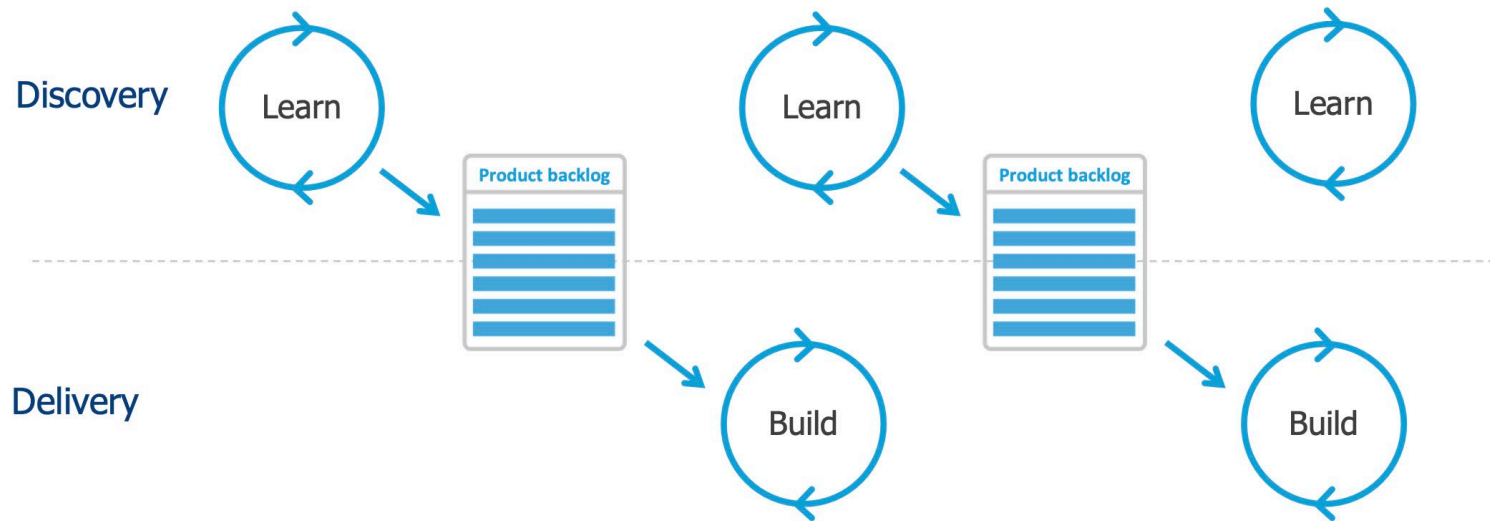
Dual Track Agile



No sequence intended – it is a list of tools for Discovery and Delivery



Dual Track Agile



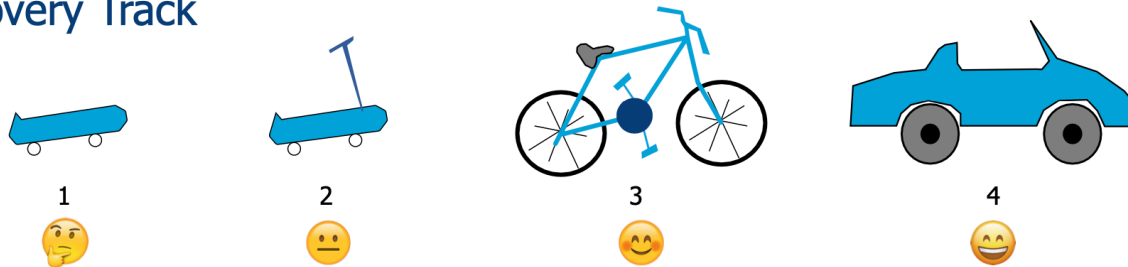
You shouldn't think of it as two processes — just two parts of one process. And, I think you're doing developers and others on the team a disservice not to involve them in discovery work. — Jeff Patton

The whole team is responsible for product outcomes, not just on-time delivery — Jeff Patton



Dual Track Agile

Discovery Track



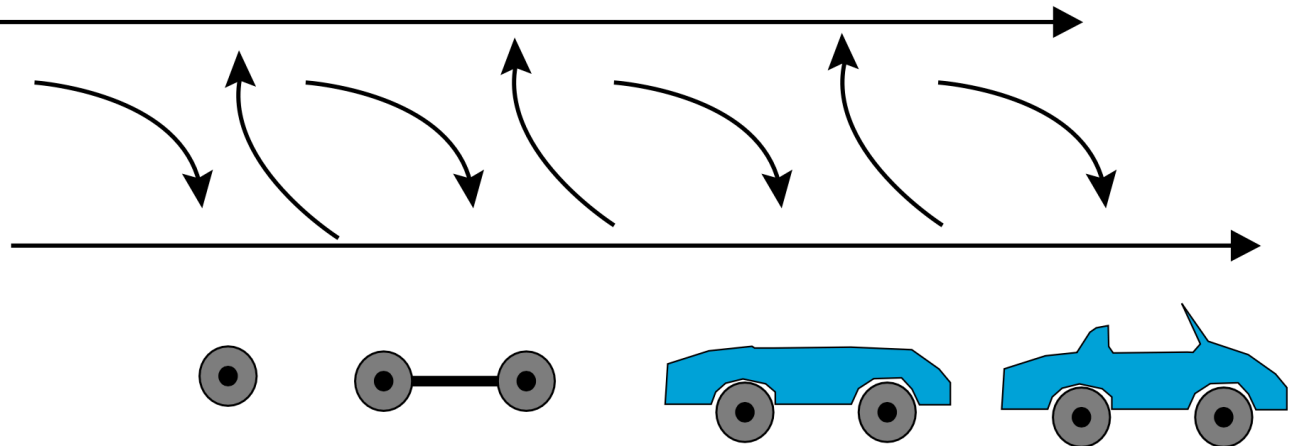
Reduce Risks Upfront

- ❖ Untested ideas Risk
- ❖ Usability Risk
- ❖ Feasibility Risk
- ❖ Viability Risk

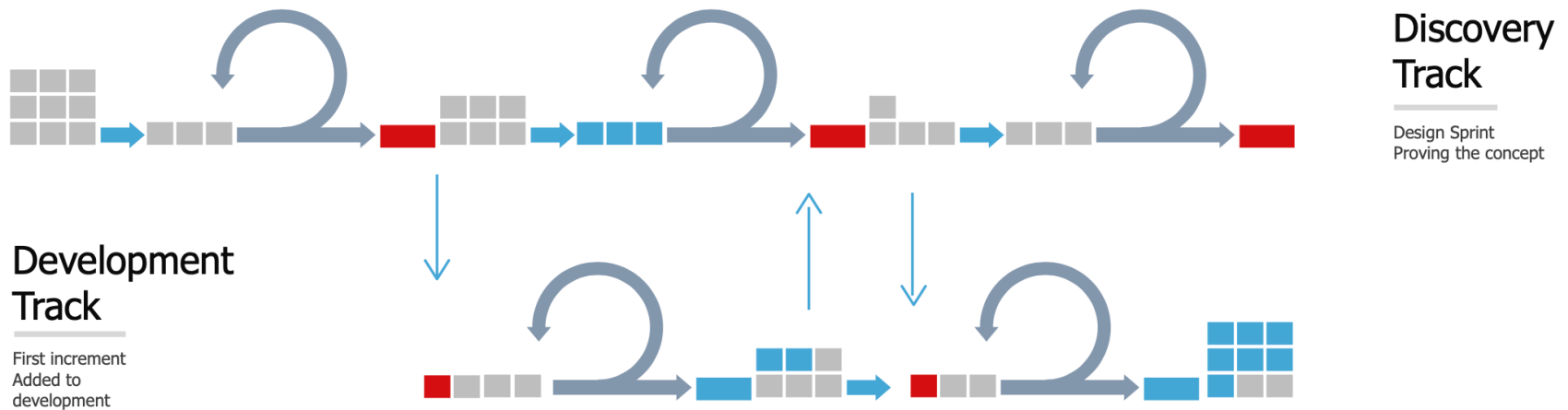
Delivery Track

Deal with

- ❖ Scalability
- ❖ Reliability
- ❖ Performance
- ❖ Maintainability



Dual Track Agile

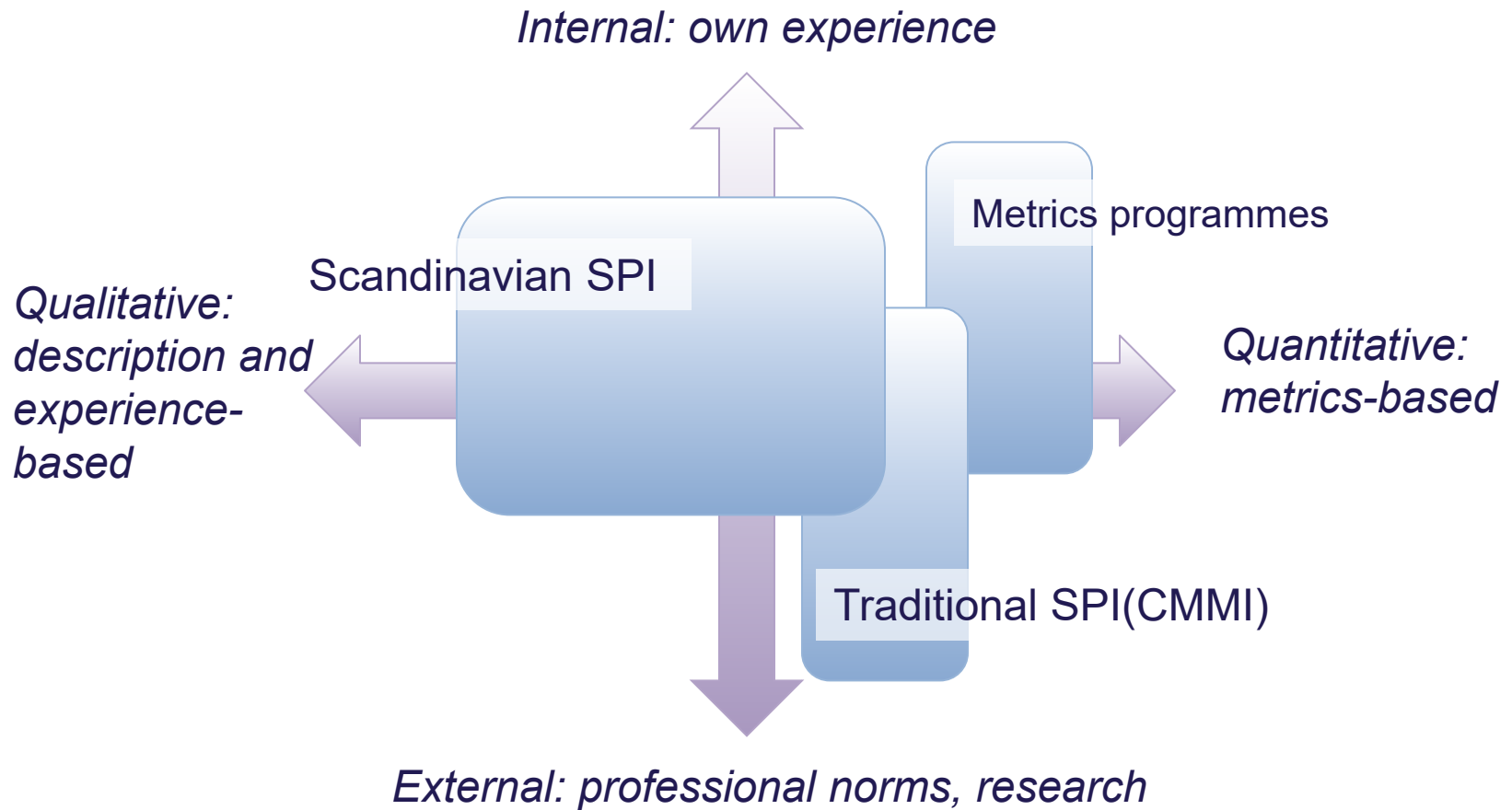


Principles

- *Release, Measure, and Learn.* — Jeff Patton
- *Two tracks, not two teams*— Jeff Patton

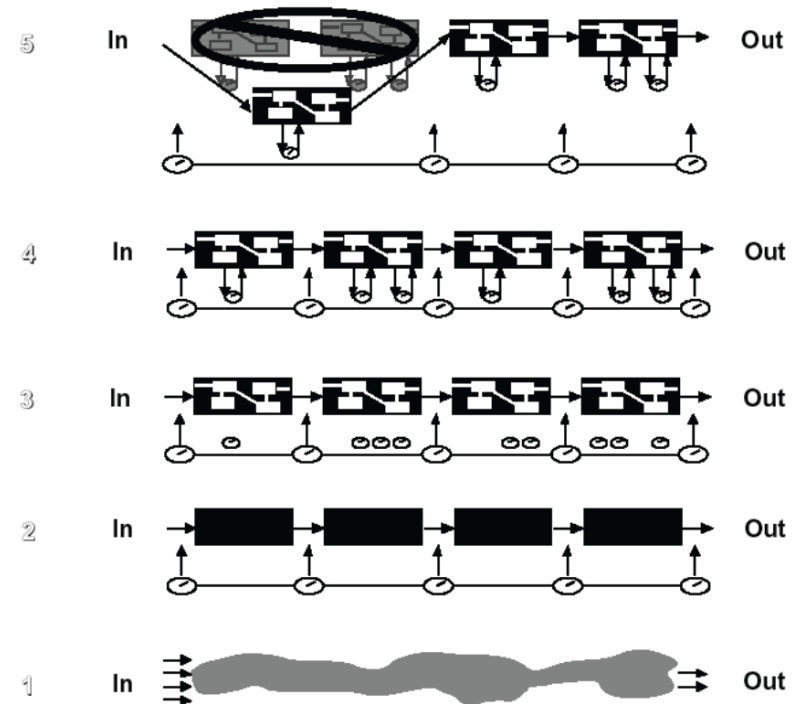
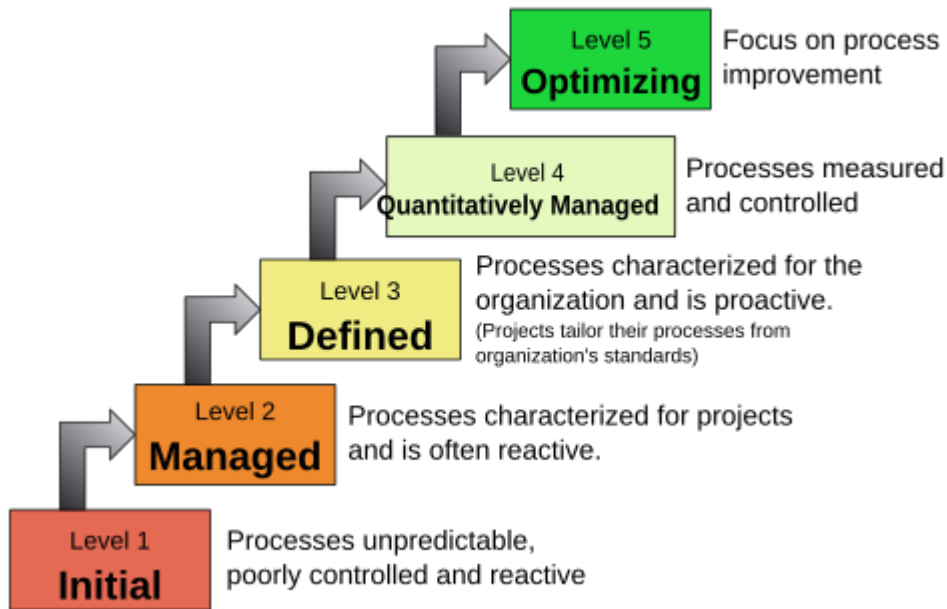


Software process improvement (SPI) styles



Capability Maturity Model Integration (CMMI)

Characteristics of the Maturity levels



The Scandinavian approach – a critique of traditional CMMI & SPI

1. Focus on problems – not on a specific solution (e.g. CMMI level 2)
2. Emphasize knowledge creation – not knowledge deployment
3. Encourage participation – not expert solution
4. Integrate leadership – do not rely on staff work
5. Plan for continuous improvement – a portfolio, not a project

Mathiassen, Nielsen & Pries-Heje 2002, Learning SPI in Practice
https://www.researchgate.net/publication/238785794_Learning_SPI_in_Practice

IMPROVING ISD AGILITY IN FAST-MOVING SOFTWARE ORGANIZATIONS

Research

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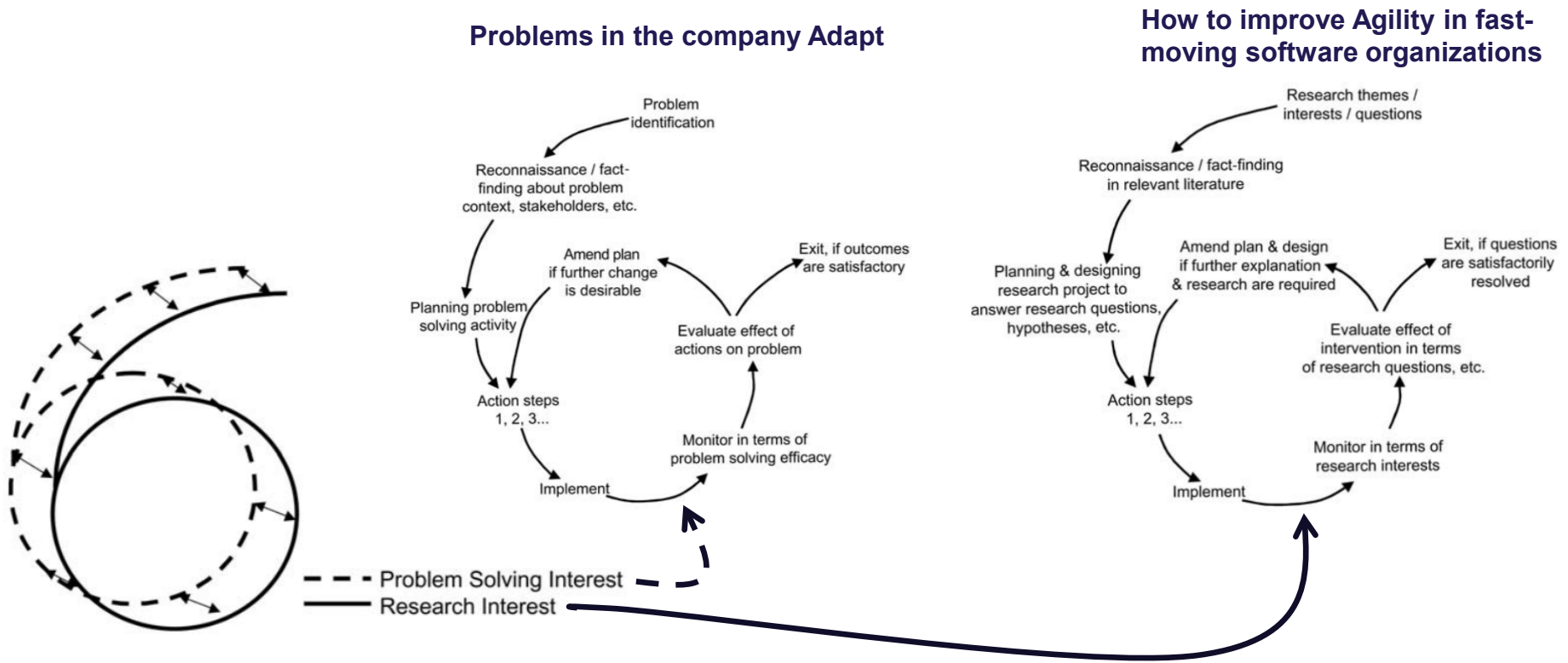
Research question:

How can Information Systems Development (ISD) agility be improved in fast-moving software organizations?



Action Research,

combines real-world problem solving with systematic learning and reflection.



McKay & Marshall (2001) The dual imperatives of action research. *Information Technology & People* 14(1), 46–59.

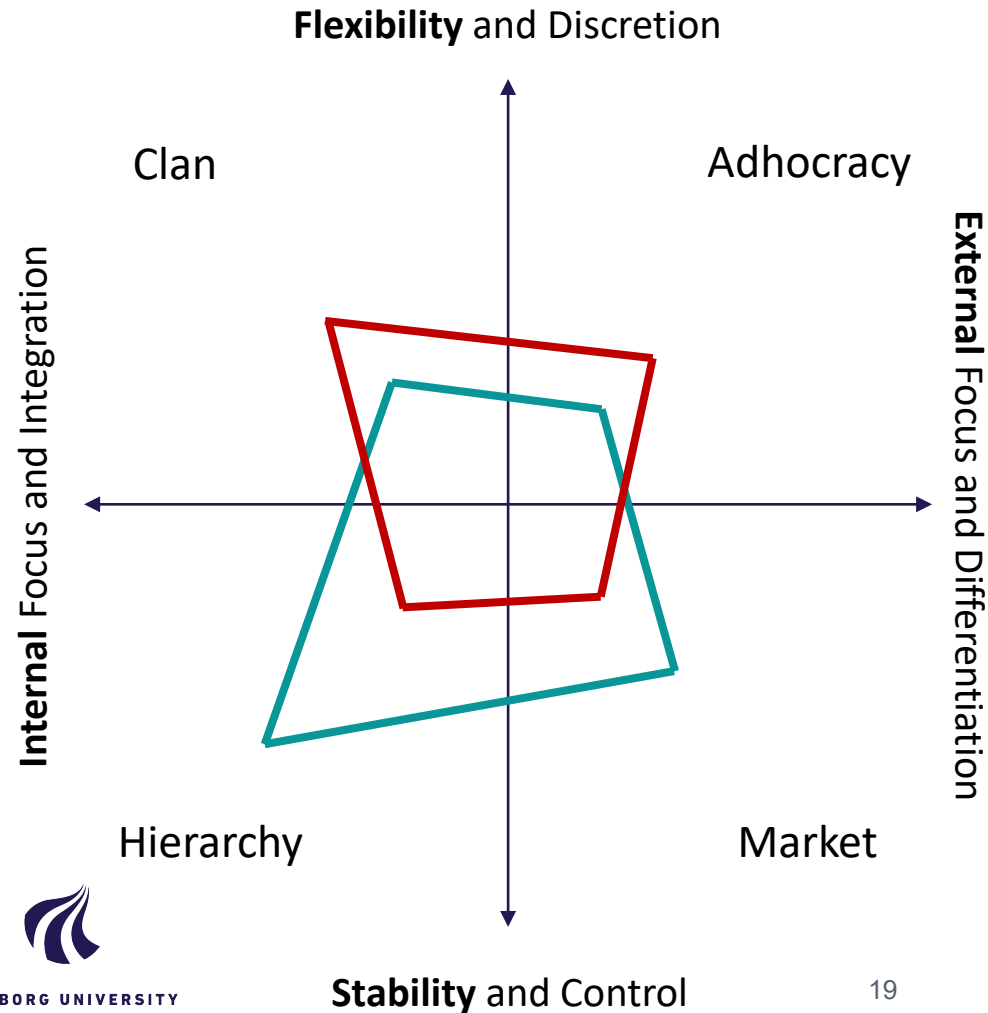


The Company Adapt

Digital agency established in 1998

- Develops web-based solutions for public and private organizations
- 60+ employees in 2014 and growing
- From medium-sized to large clients
- From fixed-price to time-and-material contracts
- From project orientation to customer orientation
- Many new hires (including project managers)

Organizational culture in Adapt vs Siemens Flow



Action

Challenges

Diverse PM responsibilities

- Developer (re)allocation
- Task management
- Customer contact
- Testing

Complex mix of contract and project models

- Waterfall – Agile
- Project — SLA
- Old – New customers
- Maintenance task — Project

Insufficient, dated, and redundant methods and tools for ISD management

- Different PM practices
- Less overview
- Redundant data entry

Intervention

1. Reformulate the ISD method to better reflect the practices and environment
 - Manage projects based on customer relationships (Project, SLA, Maintenance tasks)
 - Short descriptions based on checklists
2. Integrate ISD planning and control into one tool (*JIRA*)
 - Manage development tasks, developers, and time
 - Represent project management checklists as a JIRA workflow



Theoretical framing of Agility

1. To be agile, an ISD method component <i>must</i> contribute to one or more of the following:	(i) creation of change
	(ii) proaction in advance of change
	(iii) reaction to change
	(iv) learning from change
2. To be agile, an ISD method component <i>must</i> contribute to one or more of the following, and <i>must not</i> detract from any:	(i) perceived economy
	(ii) perceived quality
	(iii) perceived simplicity
3. To be agile, an ISD method component must be continually ready, i.e., minimal time and cost to prepare the component for use.	

Table 1. Taxonomy of ISD agility (Conboy 2009 p. 341)

Conboy, K. 2009, "Agility from First Principles: Reconstructing the Concept of Agility in Information Systems Development," *Information Systems Research* (20:3), pp. 329-354.



Evaluating contribution to agility in Adapt



		<u>(1) Define ISD management by customer relations</u>	
Taxonomy of Agility (Conboy 2009)		Project	Company
1. To be agile, an ISD method component <i>must</i> contribute to one or more of the following:	(i) creation of change		
	(ii) proaction in advance of change		
	(iii) reaction to change	Brief method descriptions contingent upon the overall task, which assumes and allows adjustments with the three ISD types as a shared staring point.	Brief method description makes itself easy to change, i.e. adding a new checkpoint or a new type of customer relationship.
	(iv) learning from change		A shared starting point in the three types helps conceptualize and discuss specific changes among the project managers and to others in the organization.
2. To be agile, an ISD method component <i>must</i> contribute to one or more of the following, and <i>must not</i> detract from any:	(i) perceived economy		
	(ii) perceived quality		
	(iii) perceived simplicity	A method scope that is lowered to their most basic practical needs specific to the different overall ISD tasks.	The method is tied to the external task rather than an internal focus on an agile versus traditional process. It also depends on and exploits the existing knowledge socialization.
3. To be agile, an ISD method component must be continually ready i.e. minimal time and cost to prepare the component for use.		The reduced method scope is easier to follow and legitimize additional questions to the experienced project managers.	The customer orientation and reduced scope ease communicating the method to existing and new project managers.

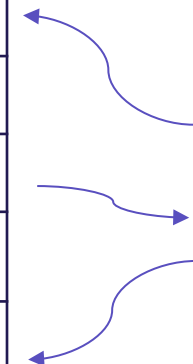
Evaluating contribution to agility in Adapt



		(2) Integrate ISD management with a tool (<i>JIRA</i>)	
Taxonomy of Agility (Conboy 2009)		Project	Company
1. To be agile, an ISD method component <i>must</i> contribute to one or more of the following:	(i) creation of change		
	(ii) proaction in advance of change	Shared overview of all the tasks and used resources helps act in advance of change.	
	(iii) reaction to change	Customers feeding tasks into <i>JIRA</i> and tracking their progress allows faster discovery and reaction to needs for change.	Checklists and templates are easier to modify (communicating change) for future projects compared to the previous underused method.
	(iv) learning from change		The checklist and templates may support shared codifications of lessons to supplement their predominantly informal learning.
2. To be agile, an ISD method component <i>must</i> contribute to one or more of the following, and <i>must not</i> detract from any:	(i) perceived economy	Time savings in setting up the ISD management environment and in having the developer team and the customers specify, manage, and coordinate tasks.	Efficient communication and integration of method changes within the ISD types and improved overview of developer resources.
	(ii) perceived quality		
	(iii) perceived simplicity		
3. To be agile, an ISD method component must be continually ready i.e. minimal time and cost to prepare the component for use.		Setting up the ISD management tool requires less effort and helps the project manager remembering and tracking her own tasks.	Setting templates and checklists, <i>JIRA</i> requires some effort but its maintenance is simpler.

A framework for improving agility

Agility		Define mgt.		Tool support	
		<i>Proj.</i>	<i>Comp.</i>	<i>Proj.</i>	<i>Comp.</i>
1. Contribute to:	(i) Create change				
	(ii) Proactive				
	(iii) React to change				
	(iv) Learn from change				
2. Contribute to and not detract from:	(i) Economy				
	(ii) Quality				
	(iii) Simplicity				
3. Continually ready:					

- 
1. Map problems
 2. Elicit overview
 3. Plan improvement
 4. Evaluate improvement



Exercise

Assignment 1: Improving agility in your project (100%)

Assess a process improvement initiative (method component) you have adopted with the table below.

Taxonomy of Agility		<<Process improvement initiative / method component>>	
		Software Development Project	Problem-based Learning Project
1. To be agile, an ISD method component <i>must</i> contribute to one or more of the following:	(i) creation of change		
	(ii) <u>proaction</u> in advance of change		
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2. To be agile, an ISD method component <i>must</i> contribute to one or more of the following, and <i>must not</i> detract from any:	(i) perceived economy		
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Lecture objectives

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